

ROSTOV-ON-DON, Russian Federation

WMO: 347300

Lat: 47.258N

Lon: 39.818E

Elev: 260

StdP: 14.56

Time Zone: 3.00 (E03)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.1	5.2	-6.0	4.0	1.8	-0.8	5.3	6.2	31.9	16.8	29.4	19.9	11.1	80	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	95.2	70.0	91.6	69.4	88.2	68.2	73.9	88.0	72.1	85.4	70.4	83.1	11.6	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.8	110.5	80.3	67.9	103.3	78.3	66.1	97.2	77.1	37.6	88.3	36.0	85.5	34.6	83.0	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.5	24.0	21.0	-3.6	99.1	7.0	3.5	-8.7	101.7	-12.8	103.7	-16.7	105.7	-21.8	108.2		
			-4.3	76.4	6.9	2.6	-9.3	78.3	-13.3	79.8	-17.2	81.2	-22.2	83.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	26.4	28.4	38.2	51.6	63.0	70.9	75.8	75.2	64.1	50.4	38.8	30.2
	DBStd	19.53	10.94	10.57	7.92	6.98	7.16	6.28	5.68	6.46	7.24	8.50	8.84	9.83
	HDD50	2835	731	605	371	63	3	0	0	0	3	100	345	614
	HDD65	6024	1196	1025	830	404	126	17	1	4	100	456	786	1079
	CDD50	3279	0	0	6	111	405	627	800	780	427	113	10	0
	CDD65	994	0	0	0	2	63	195	336	319	74	5	0	0
	CDH74	9772	0	0	0	20	561	1764	3457	3363	569	38	0	0
	CDH80	4011	0	0	0	3	136	640	1523	1547	154	8	0	0
Wind	WSAvg	9.6	10.6	11.2	11.4	10.5	8.4	8.1	8.1	8.4	9.1	9.1	9.9	10.3
Precipitation	PrecAvg	23.3	2.1	1.6	1.5	1.9	2.4	2.4	2.2	1.7	1.6	1.3	2.0	2.8
	PrecMax	31.5	5.1	3.6	3.6	5.0	5.9	5.8	6.5	4.6	6.7	4.2	5.1	5.9
	PrecMin	16.8	0.2	0.1	0.0	0.2	0.2	0.4	0.1	0.2	0.0	0.1	0.2	0.2
	PrecStd	3.7	1.3	0.9	0.8	1.0	1.3	1.3	1.5	1.2	1.3	0.9	1.3	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.8	53.9	64.2	76.8	88.9	93.5	98.9	100.5	89.5	79.2	63.0	52.0
		MCWB	44.8	47.2	51.7	59.0	70.5	69.4	72.3	70.0	69.0	64.0	54.8	47.7
	2%	DB	44.4	47.8	57.5	71.4	83.8	90.0	95.3	95.4	84.3	71.6	57.5	48.2
		MCWB	42.3	43.6	48.1	56.5	65.5	69.4	70.3	69.6	65.2	58.8	52.3	46.1
	5%	DB	41.1	44.0	53.6	67.7	80.3	87.1	91.6	91.7	80.5	66.5	53.5	44.8
		MCWB	39.6	41.1	45.7	54.7	63.8	68.1	70.2	68.4	64.0	57.6	49.7	43.0
	10%	DB	38.8	40.6	49.7	64.1	76.8	83.8	88.1	88.2	76.8	62.5	50.1	42.5
		MCWB	37.6	38.4	44.1	53.0	62.7	66.8	68.4	67.5	62.6	56.0	47.7	40.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.2	48.9	52.3	60.3	72.1	74.0	77.2	77.0	71.2	65.1	56.0	49.2
		MCDB	47.4	53.3	61.4	71.9	85.8	87.5	92.1	90.9	82.5	75.0	60.2	50.3
	2%	WB	42.0	44.1	49.6	58.0	67.9	71.8	74.2	73.6	68.1	61.3	52.8	45.9
		MCDB	43.6	46.4	56.3	68.3	79.4	83.9	88.3	87.9	79.0	68.7	57.3	47.8
	5%	WB	39.4	41.0	46.7	56.0	65.6	70.2	72.5	71.5	66.0	59.0	50.2	43.0
		MCDB	40.5	43.0	51.8	65.0	76.5	82.3	86.1	84.5	76.3	64.6	52.6	44.4
	10%	WB	37.1	37.9	44.2	54.2	63.5	68.6	70.5	69.7	64.1	56.4	47.8	40.6
		MCDB	38.1	39.6	48.6	62.3	73.6	80.1	83.4	83.0	73.8	61.8	50.0	41.8
Mean Daily Temperature Range	5% DB	MDBR	9.0	11.1	13.5	17.5	18.7	18.7	19.5	20.9	19.1	15.1	10.8	9.1
		MCDBR	9.9	14.6	20.6	22.8	23.2	22.6	23.5	24.7	23.6	21.0	15.1	11.7
		MCWBR	8.9	11.1	12.8	12.1	11.3	9.8	9.8	10.2	11.1	12.6	11.0	10.1
	5% WB	MCDBR	9.4	13.2	18.4	19.9	21.1	19.8	20.8	21.5	21.2	19.3	14.0	11.0
		MCWBR	8.7	10.7	12.2	11.2	11.4	10.2	10.5	11.1	11.3	12.3	10.9	9.9
Clear-Sky Solar Irradiance	taub	0.342	0.375	0.412	0.457	0.419	0.424	0.448	0.447	0.416	0.386	0.355	0.346	
	taud	2.221	2.125	2.127	2.057	2.255	2.269	2.212	2.208	2.262	2.305	2.312	2.236	
	Ebn at Noon	225	241	252	252	269	267	259	253	249	238	220	207	
	Edh at Noon	26	36	42	49	42	42	43	42	36	30	24	23	
All-Sky Solar Radiation	RadAvg	311	592	933	1459	1906	2051	2010	1820	1312	766	377	239	
	RadStd	66	99	100	145	184	155	107	121	110	106	69	37	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+128
(47)	Regional (9 neighbors)	+1.17	N/A	N/A	N/A	N/A	N/A	N/A	-270	+236	+167

Nomenclature: See separate page